Pre-Calculus Concepts

ANSWER KEY

Section 1: Sequences & Partial Sums

1 In an arithmetic sequence, the 4th term is 19 and the 11th term is 54. What is the first term a_1?

a. 4 ✓

b. 5

c. 6

d. 7

Common difference $d = \frac{a_{11} - a_4}{11 - 4} = \frac{54 - 19}{7} = \frac{35}{7} = 5$. Then $a_1 = a_4 - 3d = 19 - 3(5) = 19 - 15 = 4$.

2 What is the sum of the first 25 terms of the arithmetic series $3 + 7 + 11 + 15 + \ldots$?

a. $1,250$

b. $1,275$ ✓

c. $1,300$

d. $1,325$

$a_1 = 3$, $d = 4$. Term $a_{25} = 3 + 24(4) = 99$. Sum $S_{25} = \frac{25}{2}(3 + 99) = \frac{25}{2}(102) = 25 \times 51 = 1,275$.

3 What is the sum of the infinite geometric series $18 - 12 + 8 - \frac{16}{3} + \ldots$?

a. $\frac{54}{5}$ ✓

b. $\frac{36}{5}$

c. 10

d. 12

First term $a = 18$, common ratio $r = -\frac{12}{18} = -\frac{2}{3}$. Since $|r| < 1$, sum $S = \frac{a}{1 - r} = \frac{18}{1 - (-2/3)} = \frac{18}{5/3} = \frac{54}{5}$.

Section 2: Binomial Expansion & Pascal's Triangle

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4 What is the coefficient of x^3y^2 in the binomial expansion of $(2x - y)^5$?

a. 40

b. -40

c. 80 ✓

d. -80

Binomial term: $\binom{5}{2}(2x)^3(-y)^2 = 10 \times (8x^3) \times y^2 = 80x^3y^2$. The coefficient is 80.

5 What is the sum of all coefficients in the expansion of $(3x - 2y)^6$?

a. 1 ✓

b. 64

c. 729

d. 0

To find the sum of coefficients, substitute $x = 1$ and $y = 1$: $(3(1) - 2(1))^6 = (1)^6 = 1$.

6 What is the 4th term in the expansion of $(x + 2)^5$ arranged in descending powers of x?

a. $40x^2$

b. $80x^2$ ✓

c. $20x^3$

d. $10x^3$

4th term has $k = 3$: $\binom{5}{3}x^{5-3}(2)^3 = 10x^2(8) = 80x^2$.

Section 3: Intuitive Limits & Continuity

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7 Evaluate the algebraic limit: $\lim_{x\to 3} \frac{x^2-9}{x^2-2x-3}$.

a. $\frac{3}{2}$ ✓

b. $\frac{6}{5}$

c. $\frac{3}{4}$

d. Does not exist

Factor numerator and denominator: $\lim_{x\to 3} \frac{(x-3)(x+3)}{(x-3)(x+1)} = \lim_{x\to 3} \frac{x+3}{x+1} = \frac{3+3}{3+1} = \frac{6}{4} = \frac{3}{2}$.

8 Evaluate the limit at infinity: $\lim_{x\to\infty} \frac{4x^3-5x+2}{2x^3+7x^2-1}$.

a. 0

b. 2 ✓

c. 4

d. ∞

Divide all terms by the highest power x^3: $\lim_{x\to\infty} \frac{4-5/x^2+2/x^3}{2+7/x-1/x^3} = \frac{4}{2} = 2$.

9 Evaluate the limit involving radicals: $\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$.

a. $\frac{1}{4}$ ✓

b. $\frac{1}{2}$

c. 1

d. 0

Rationalize the numerator: $\frac{(\sqrt{x+4}-2)(\sqrt{x+4}+2)}{x(\sqrt{x+4}+2)} = \frac{x}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2}$. As $x\to 0$, this becomes $\frac{1}{2+2} = \frac{1}{4}$.

Section 4: Difference Quotients & Tangents

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10 For $f(x) = 3x^2 - 5x$, simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$.

a. $6x + 3h - 5$ ✓

b. $6x - 5$

c. $6x + 3h$

d. $3x + 3h - 5$

$f(x+h) = 3(x+h)^2 - 5(x+h) = 3x^2 + 6xh + 3h^2 - 5x - 5h$. Subtract $f(x)$: $6xh + 3h^2 - 5h$. Divide by h: $6x + 3h - 5$.

11 What is the slope of the tangent line to the curve $y = x^2 - 4x$ at $x = 3$?

a. 2 ✓

b. 3

c. 4

d. 6

The derivative is $y' = 2x - 4$. At $x = 3$: $y'(3) = 2(3) - 4 = 2$.

12 At what value of x does the function $f(x) = -2x^2 + 12x - 5$ achieve its maximum value?

a. 2

b. 3 ✓

c. 4

d. 6

Vertex occurs at $x = -\frac{b}{2a} = -\frac{12}{2(-2)} = 3$.

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